Vishnu Kumar

Final year CSE student at NIT Patna, graduating in 2027. I mostly work in C++ and Python, spend most of my time on DSA and problem solving, and build web apps.

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EDUCATION

National Institute of Technology, Patna — Patna, Bihar

Dual Degree (B.Tech + M.Tech) in CSE with specialization in Data Science – ***CGPA: 8.19/10*** — *2022 – 2027 (Expected)*

RSM Inter College — Sonbhadra, UP

Intermediate (UP Board) – ***Percentage: 81.66%*** — *2020*

HV Inter College — Sonbhadra, UP

High School (UP Board) – ***Percentage: 89.33%*** — *2018*

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript, SQL
Tools: Git, GitHub, VS Code
Technologies: React.js, Node.js, Express.js, REST APIs, Tailwind CSS, Bootstrap
Databases: MySQL, MongoDB
Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming, Computer Networks, Machine Learning, Deep Learning, Natural Language Processing, Data Visualization Techniques

PROJECTS

HouseHelp Connect live | *React.js, Node.js, Express.js, MongoDB, Tailwind* — Aug 2026 – Present

- A **marketplace for home services** — households find, book and pay local workers (cleaning, cooking, plumbing, childcare and similar work), and workers find jobs on it. Live on **Vercel** with a **Node/Express** backend and **MongoDB Atlas**.
- Built the main flows end to end: **search by service and city**, worker profiles, **bookings**, messaging, reviews, and a demo **payment** flow.
- Login is **role based** (client and worker) with **JWT** authentication and **bcrypt** password hashing. Workers go through **Aadhaar/PAN verification** before they appear in search.
- The frontend and backend are separate apps connected through **REST APIs**, with **CORS** handled for the cross-domain setup. Also handled the free-tier cold start with a **"server is waking up" banner** and automatic retry.

Extractive Question Answering System code | *Python, PyTorch, Transformers* — Dec 2025

- Developed an **extractive question answering** system using **DistilBERT** fine-tuned on **SQuAD**, implementing **character-to-token span alignment** and **sliding-window tokenization**.
- Achieved **70.86 EM** and **84.10 F1** on the SQuAD validation set after GPU fine-tuning.

Next Basket Item Prediction code | *Python, Pandas, Scikit-learn, PyTorch* — Mar 2024

- Built a **hybrid recommendation system** combining **positive and negative association-rule mining** with **contrastive learning** to generate reusable **item embeddings**.
- Integrated the learned item embeddings into a **Transformer-based SASRec model** for sequential next-item prediction, and evaluated it on the **Dunnhumby** retail transaction dataset using **Recall@10** and **NDCG@10**.

ACHIEVEMENTS

- **LeetCode: Knight** — Max Rating: **1898**; solved **1K+ DSA and SQL problems**, covering advanced data structures, dynamic programming, and database querying.
- **CodeChef: 3-Star** (1610) • **Codeforces: Pupil** (1244) • **GeeksforGeeks**: solved **480+ DSA problems**.
- **1st Place, College Robotics Competition (2nd Year)**: Designed and built a high-accuracy line-following robot using an **Arduino UNO** microcontroller and **infrared (IR) sensors**.

EXTRACURRICULAR ACTIVITIES

- **Sports:** Cricket and badminton — teamwork and regular practice.